30 Days Python Challenge (Beginner → Job Ready)

🔹 Week 1: Basics (Foundation Strong)

Goal: Syntax + logic building

Day 1: Print, variables, data types
👉 Program: Hello World, user input, type checking

Day 2: Operators
👉 Arithmetic, comparison, logical

Day 3: Conditional statements
👉 If-else, nested if
👉 Task: Check even/odd, largest of 3 numbers

Day 4: Loops
👉 for loop, while loop
👉 Task: Print tables, sum of numbers

Day 5: Break, Continue, Pass
👉 Task: Prime number

Day 6: Functions
👉 Create reusable functions
👉 Task: factorial function

Day 7: Revision + Mini Test
👉 Build: Simple calculator

🔹 Week 2: Data Structures (Important for Interviews)

Goal: Work with real data

Day 8: Strings
👉 Reverse, palindrome, slicing

Day 9: String problems
👉 Count vowels, frequency

Day 10: Lists
👉 Add/remove/sort

Day 11: List problems
👉 Second largest, duplicates

Day 12: Tuples & Sets
👉 Unique elements

Day 13: Dictionaries ⭐ (VERY IMPORTANT)
👉 Key-value operations
👉 Task: Count word frequency

Day 14: Revision + Practice

🔹 Week 3: Intermediate + Real Use

Goal: Automation mindset

Day 15: File Handling
👉 Read/write logs

Day 16: Exception Handling
👉 Try-except (important for support roles)

Day 17: Modules & Libraries
👉 math, random

Day 18: Date & Time
👉 Useful for logs

Day 19: Working with OS
👉 File/folder operations

Day 20: Regular Expressions (🔥 must for support)
👉 Log parsing

Day 21: Mini Project
👉 Log Analyzer (read file → find ERROR lines)

🔹 Week 4: Job-Ready Skills (🔥 Important for YOU)

Goal: AWS + Support + Automation

Day 22: API Basics
👉 Use `requests` module

Day 23: JSON Handling
👉 Parse API response

Day 24: Automation Script
👉 Restart service / check logs

Day 25: Database (SQLite basics)
👉 Insert, fetch data

Day 26: CLI Tool
👉 Build command-line script

Day 27: AWS Intro with Python
👉 Install & use Boto3

Day 28: AWS Script
👉 List S3 buckets, upload file

Day 29: Final Project
👉 "System Monitoring Script"

- Check CPU/logs
- Send alert

Day 30: Revision + Mock Interview
👉 Explain your scripts like real interview

💡 Real Projects You'll Build

✓ Calculator
✓ Log Analyzer (VERY important for your role)
✓ API data fetcher

✓ AWS S3 automation script
✓ System monitoring tool

🎯 How YOU should follow this (very important)

- ⏱️ Daily: 1–2 hours
- ✍️ Write code (don't just read)
- 📁 Save in GitHub (helps for jobs)
- 🔁 Revise every Sunday

🔥 Bonus (For Your Interviews)

Focus extra on:

- File handling (log analysis)
- Exception handling
- Regex (log parsing)
- AWS + Python (Boto3)